

Tarpons and eels

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Elopomorpha
ORDERS	4
FAMILIES	24
SPECIES	About 1,000

This group consists of 3 main lines: eels, tarpons, and halosaurs. Although they look and behave differently as adults, they all start life as a transparent larva (known as a leptocephalus larva). These leaflike larvae may drift in ocean currents for several years before reaching adulthood. Most eels are marine, but a few live mainly in freshwater and return to the sea to breed. Tarpons are largely confined to coastal areas of the tropics, while halosaurs occur in the deep sea worldwide, down to depths of at least 6,000ft (1,800m).

Anatomy

Eels are elongated and snakelike. The spinal column consists of more than 100 vertebrae, which makes the body very flexible. Long dorsal and anal fins may extend most of the length of the body, and the pectoral and tail fins are often very small or lost entirely. There are no pelvic fins. Most eels are scaleless, but some species have fine scales embedded in their skin. Tarpons look more like large herring than eels. Their bodies are deeper and less elongated, with large, reflective scales, and well-developed and differentiated fins. Halosaurs, however, are more eellike in shape, with a large

head that tapers into a tubular body and elongated tail. They are covered in small scales and have reduced fins, some of which have sharp spines.

Breeding and migration

The breeding cycles of some eels involve remarkable migrations. Most marine eels spawn near where they live, but freshwater eels and some marine species have separate spawning grounds, usually far out to sea. For example, North American and European eels, both of which live in freshwater lakes and rivers, travel over 4,000 miles (6,400km) to different parts of the Sargasso Sea in the western Atlantic. It is thought that during this journey, which takes 4–7 months, they do not eat. Instead, they survive by consuming stored fat and muscle tissue. They may also absorb nutrients through the lining of their mouth. Spawning occurs in deep

water, after which the adults die. The leptocephalus larvae are carried passively toward coastal waters by ocean currents. After about 1–3 years, they develop fins, scales, and pigmentation before beginning their journey upriver to the areas where they grow as adults.

Hunting and feeding

Tarpons, halosaurs, and eels are all predators. The fast-swimming tarpons use speed to catch other fishes. Eels are more likely to ambush their prey, using rocks or reef crevices for cover. Many eels have long, sharp teeth for seizing prey; others have thickened teeth for crushing shells. Some species search the bottom for buried prey or scavenge on dead animals. Deep-sea gulper eels have gigantic mouths and distensible stomachs, and can eat prey much larger than themselves. Halosaurs feed on invertebrates.



HUNTING
Eels catch prey in different ways but many ambush prey rather than chase it, as tarpons do. Moray eels (shown here) often hide in crevices, from which they launch surprise attacks when suitable prey is within reach. However, they also emerge at night and actively hunt their prey. The wide mouth is armed with sharp teeth, making it difficult for prey to escape once it has been caught.